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Combinatorial Optimization Using Tabu Search

Homework 5: 15-Department Quadratic Assignment Problem

1. Introduction

In this project, I solved the Quadratic Assignment Problem (QAP) using a tabu search (TS) algorithm. This algorithm developed by Fred Glover in the 1970s [1]. TS is a meta-heuristic designed for combinatorial optimization problems. The algorithm, in its canonical form, that explores the whole neighborhood of a single solution in each iteration. This making it a single-solution algorithm rather than population-based. Having defined and explored the whole neighborhood with a specific move operator and encoding, then the algorithm proceeds to make the best possible move [2].

The main characteristic of this algorithm is the tabu list that restricts the moves. To avoid a greedy behavior, tabu search saves movements that would make it go back on the search path. This way the algorithm does not loop around local optimums in its iterations. In its canonical form, the tabu list has a fixed size and the restrictions of movements saved in this list expire given a certain number of executed iterations. This aspect of the algorithm is often referred to as short-term memory. The degree of restrictions to the moves not only depends on the size of the tabu list but also on its entries. Depending on how the movement is saved, so the possibilities of the algorithm could be greatly altered [1,2].

A common extension of this algorithm is the aspiration criterion. This characteristic, often referred to as long-term memory, saves a certain aspect of the search and gives the algorithm the possibility to make a restricted move if the criterion is met. The usual selection of the aspiration criteria is the best solution so far.

For this homework, the problem to solve is the quadratic assignment problem with 15 departments (Nugent Nug15 benchmark). this problem consists of locating 15 departments in a grid of three rows and five columns, what we should minimizing the total flow cost between all pairs of departments based on their assigned locations [3].

Considering all this, in this report an analysis on the initial solutions is explored. Then, different levels for the tabu size are executed. Finally, several variations of the algorithm are implemented, including aspiration criteria, partial neighborhood exploration, and diversification strategies, and conclusions are described for the executions of this homework [4].

2. Methodology, Definitions and Assumptions

2.1. Problem definition

The optimization target is the 15-department Quadratic Assignment Problem from Nugent et al. (1968), known as Nug15 [3]. Fifteen departments must be assigned to 15 locations arranged in a 3 * 5 grid. The objective is to minimizing the total flow cost, which is calculated as the sum of flow and distance for all pairs of departments. Both flow and distance matrices are symmetric. The known optimal solution has a cost of 1150 (using the symmetric flow × distance formulation).

2.2. Solution encoding

Each solution is encoded as a permutation of integers [0, 1, ..., 14]. where the index represents the department and the value at that index represents the assigned location. This encoding chosed because it is a simple way to represent a solution in a symmetric location problem like the QAP.

2.3. Move operator and neighborhood

The move operator selected is a pairwise swap: two departments exchange their assigned locations. This swap ensures that the resulting solution is always feasible. Also, The neighborhood defined by this move operator consists of all C(15, 2) = 105 possible pairwise swaps from the current solution. For the partial neighborhood variant, a random 50% subset of these 105 swaps is sampled at each iteration.

2.4. Tabu list entry

The tabu entry is defined by the pair of department indices (i, j) involved in the swap. When a swap of departments i and j is performed, the pair (i, j) is added to the tabu list, that preventing the algorithm from swapping these same two departments again until the entry expires. This is a moderately restrictive tabu definition, that it prevents reversing the exact move but does not prevent the individual departments from participating in other swaps.

2.5. Stopping criterion

All runs use a fixed stopping criterion of 500 iterations. This value was selected to ensure sufficiently long runs for the algorithm to converge while allowing exploration of diversification mechanisms such as random restarts.

2.6. Experimental design

To evaluate the progressive improvement of TS design choices, seven experiments are conducted with the same five random seeds (42, 123, 256, 789, 1024) for a total of 35 runs. The details are summarized in the following table.

Table 1: Detail of the experiments (35 total runs across 7 TS variants)

Experiment	**Tabu Size**	**Neighborhood**	**Aspiration**	**Diversification**	**Runs**
E1: Basic TS	10	Complete	None	None	5
E2: Small tabu	5	Complete	None	None	5
E3: Large tabu	15	Complete	None	None	5
E4: Dynamic tabu	5–15 (random)	Complete	None	None	5
E5: Aspiration	10	Complete	Best-so-far	None	5
E6: Partial NH	10	50% random	Best-so-far	None	5
E7: Diversification	10	Complete	Best-so-far	Random restart	5

3. Results and Discussion

3.1. Preliminary runs (Experiment 1: Basic TS)

3.1.1. Parameter definition

The preliminary run's objective is to test the algorithm's sensitivity to the randomness of the initial solution. This parameters for the execution of the algorithm is the only source of variation in the canonical form of the algorithms. Once defined the initial solution, the search that these canonical tabu search executes is completely deterministic. So, it is important to analyze how this randomness affects the algorithm. For this purpose, a preliminary parameter for the tabu length was selected with the class recommendation of a value between 7 and 20. With this in mind, a value of 10 was chosen.

3.1.2. Results

The following graphs show the near optimums found in the execution of the canonical tabu search with a tabu list size of 10 for 5 different seeds.

Table 2: Experiment 1 results — Basic TS with 5 seeds

Seed	Init Cost	Best Cost	Gap from Optimal	Runtime (s)
42	1646	1150	0	0.23
123	1648	1150	0	0.23
256	1616	1160	10	0.24
789	1602	1152	2	0.24
1024	1648	1150	0	0.23

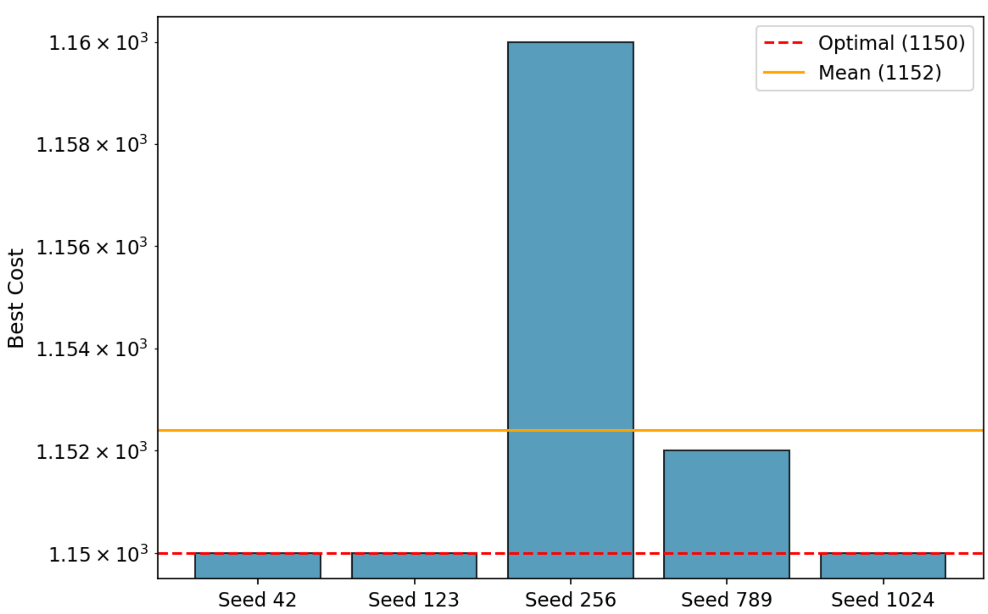

Figure 1: Near-optimums reported by seed for the preliminary runs

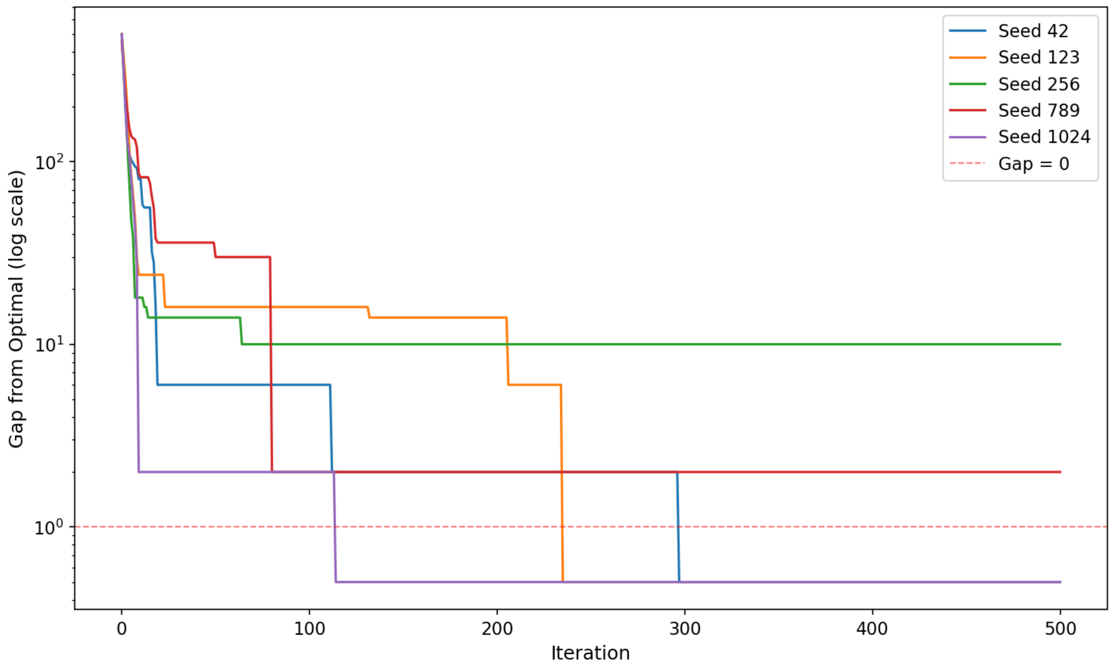

Figure 2: Convergence gap from optimal by seed (Experiment 1, log scale)

3.1.3. Conclusions of the preliminary runs

From the previous results, it can be concluded that the basic tabu search with a tabu size of 10 performs very well on this problem. Out of 5 seeds, 3 reached the known global optimum of 1150. The mean best cost across the 5 runs was 1152.4. The algorithm converges rapidly, typically reaching near-optimal solutions within the first 50–80 iterations.

The initial solution influences which sections of the search space the algorithm explores. However, since the canonical tabu search is completely deterministic after the initial solution is set, restarting from a different starting point can be a good strategy for improving results.

3.2. Tabu length tuning runs

3.2.1. Definitions

The main parameter to be tuned in tabu search (other than important definitions as the aspiration criterion) is the length of the tabu list. This parameter has a great influence on how much the algorithm will restrict its search. If a greater length is selected, then the restricted moves will last longer in the operation of the tabu search and will have more chances to defy improving moves. If the tabu length is shorter, so then the algorithm will restrict fewer moves and will act more like a greedy search.

To explore the influence of this parameter, three fixed levels (5, 10, 15) and a dynamic variant were tested for the same 5 seeds. The dynamic implementation changes the tabu size every 20 iteration by sampling a new integers from uniform (5, 15).

Table 3: Levels for the tabu length testing

Implementation	Values	Justification
Fixed length	5, 10, 15	Range considering class

		recommendation interval (7–20)
Dynamic length	Uniform(5, 15)	Random uniform within the same range, changed every 20 iterations

3.2.2. Results

For each of the four options to define the length of the tabu list, five seeds were executed. The results of these runs are shown in the boxplots of the following graph.

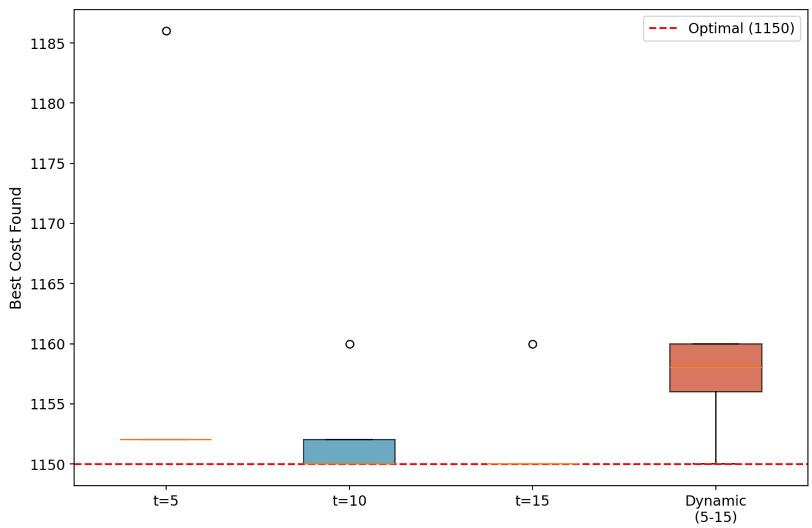

Figure 3: Boxplots of the different options for tabu list length

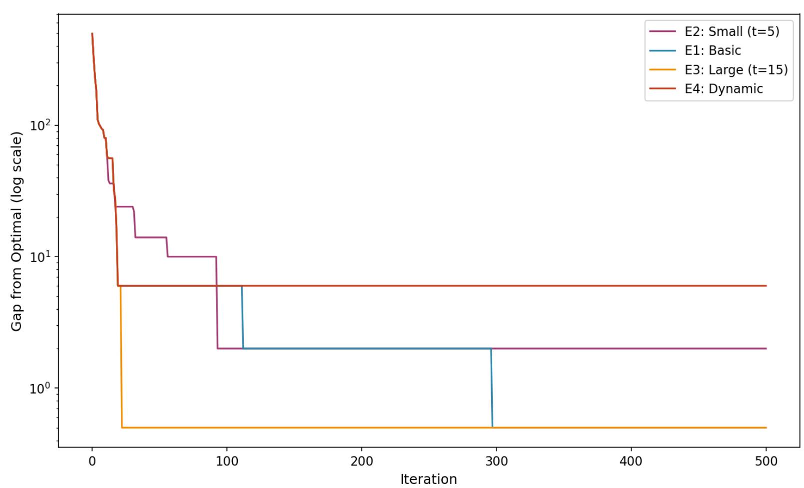

Figure 4: Convergence gap comparison of tabu list sizes

Table 4: Average results of tabu list length experimentation

Tabu Size	**Mean Best Cost**	**Std Dev**	**Min**	**Max**

5	1158.8	15.2	1152	1186
10	1152.4	4.3	1150	1160
15	1152.0	4.5	1150	1160
Dynamic (5–15)	1156.8	4.1	1150	1160

3.2.3. Conclusions of the tabu length tuning runs

The previous results show that the larger tabu list size (t=15) achieved the best performance, that 4 out of 5 seeds reached the global optimum and the lowest mean cost. The smaller tabu list (t=5) performed worst, that as the reduced restriction led to more greedy behavior and susceptibility to cycling near local optima. So, the dynamic tabu list performed comparably to the fixed sizes. So this suggesting that introducing randomness in the tabu size does not provide a significant advantage for this problem instance.

Considering these results, the value of 10 for the tabu list length was chosen as the base for subsequent experiments. This provides a good balance between restriction and flexibility.

3.3. Aspiration criteria, partial neighborhood and diversification runs

3.3.1. Definitions

For this section, three different variations of the algorithm are tested; the implementation of an aspiration criterion, the exploration of less than the full neighborhood for each iteration, and the incorporation of diversification mechanisms in the search.

Also, the aspiration criterion implemented is best solution so far. This way the algorithm will ignore the tabu list only when the possible move results in a better improvement in the objective function, than all the previous evaluated values.

For the partial neighborhood evaluation, only half of the possible neighbors were evaluated (approximately 53 out of 105 swaps). This was implemented by randomly sampling 50% of the swap pairs at each iteration. This way, a random operator to select which permutation to explore was incorporated.

Finally, for the diversification strategy, random restart was implemented. This strategy activates when no improvement is achieved in 30 consecutive iterations. Upon activation, the algorithm generates a completely new random permutation, and clears the tabu list. Thus allowing the search to explore an entirely new region of the search space.

3.3.2. Results

Aspiration criteria:

For the aspiration criteria, the same 5 seeds were used with a tabu size of 10. The boxplots compare the results with and without aspiration.

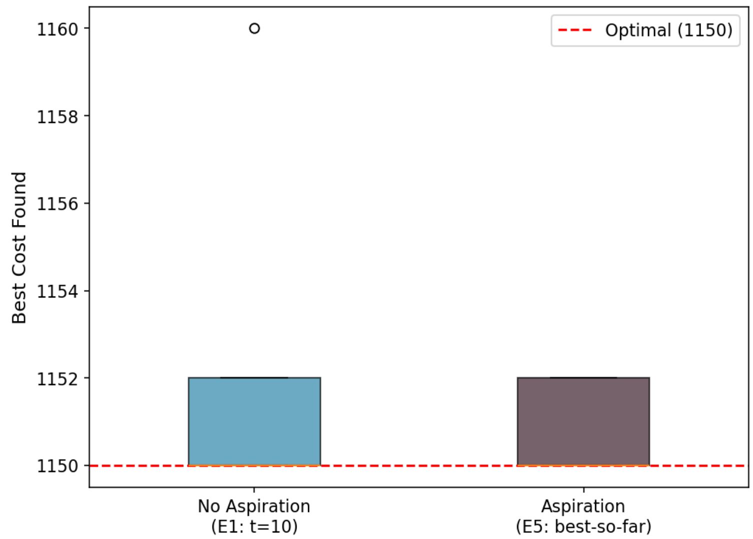

Figure 5: Boxplots with and without aspiration criteria

Table 5: Consolidated results of the aspiration criteria runs

Configuration	**Mean**	**Std Dev**	**Min**	**Max**
With Aspiration (E5)	1150.8	1.1	1150	1152
Without Aspiration (E1)	1152.4	4.3	1150	1160

The aspiration criterion improved performance, that achieving a lower mean cost of 1150.8 compared to 1152.4 without aspiration. The standard deviation also decreased notably. This indicating more consistent results. The aspiration mechanism allows the algorithm to override the tabu restriction when a genuinely superior solution is found, which effectively preventing the algorithm from missing good moves that happen to be tabu.

Partial neighborhood:

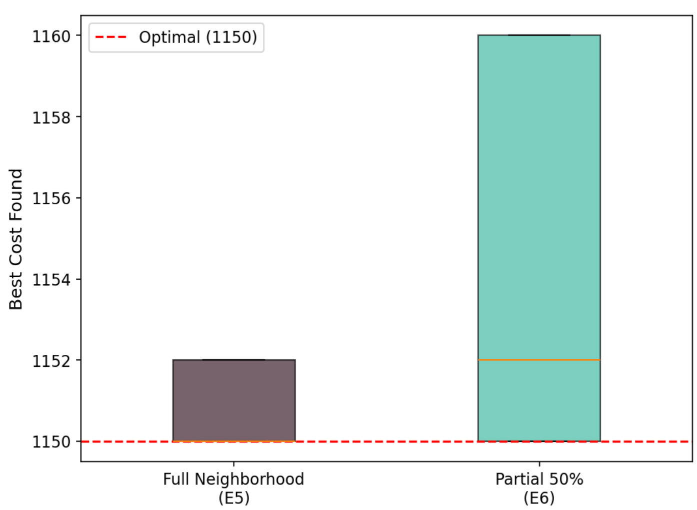

Figure 6: Boxplots with and without partial neighborhood

The partial neighborhood strategy produced a mean cost of 1154.4 compared to 1150.8 for the full neighborhood with aspiration. While the partial neighborhood introduces randomness and reduces computation time, it slightly increased the mean cost. The stochastic element means some good moves may be missed in a given iteration, but this also adds diversification by preventing the algorithm from following the same deterministic path.

Diversification strategies:

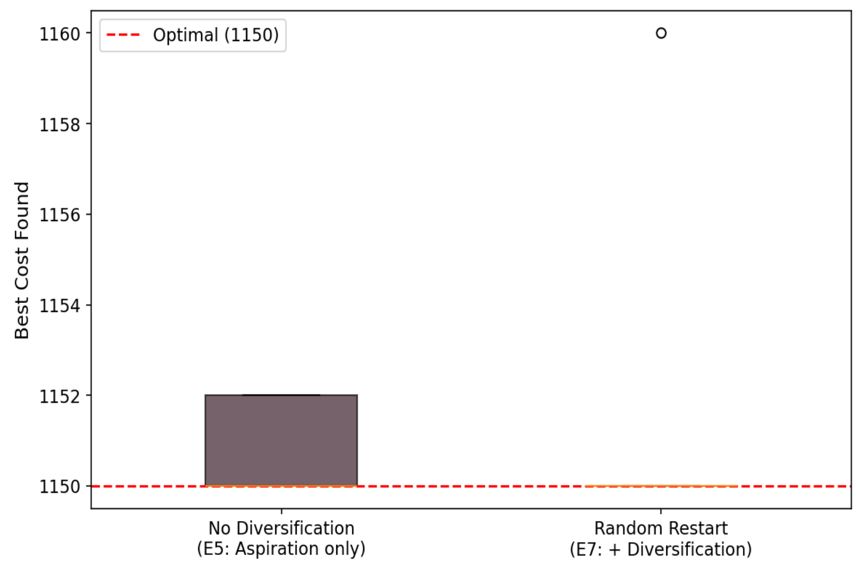

Figure 7: Boxplots with and without diversification strategies

The random restart diversification strategy performed well, with 4 out of 5 seeds reaching the global optimum of 1150, and a mean cost of 1152.0. Also, the random restart mechanism is effective because when the algorithm stagnates, it jumps to a completely new region of the search space, increasing the chance of finding the global optimum.

3.3.3. Conclusions of the aspiration, partial neighborhood and diversification runs

As a conclusion for this section, several observations can be made. The aspiration criterion improved mean performance and consistency, which making the search less sensitive to the starting solution. The partial neighborhood adds stochasticity which can help diversification but may miss good moves. The random restart diversification is the most effective strategy for escaping local optima and exploring new regions.

3.4. Comparison across all experiments

Figure 8 presents boxplots comparing the best cost achieved by each TS variant across all 35 runs. The optimal cost of 1150 is shown as a red dashed line.

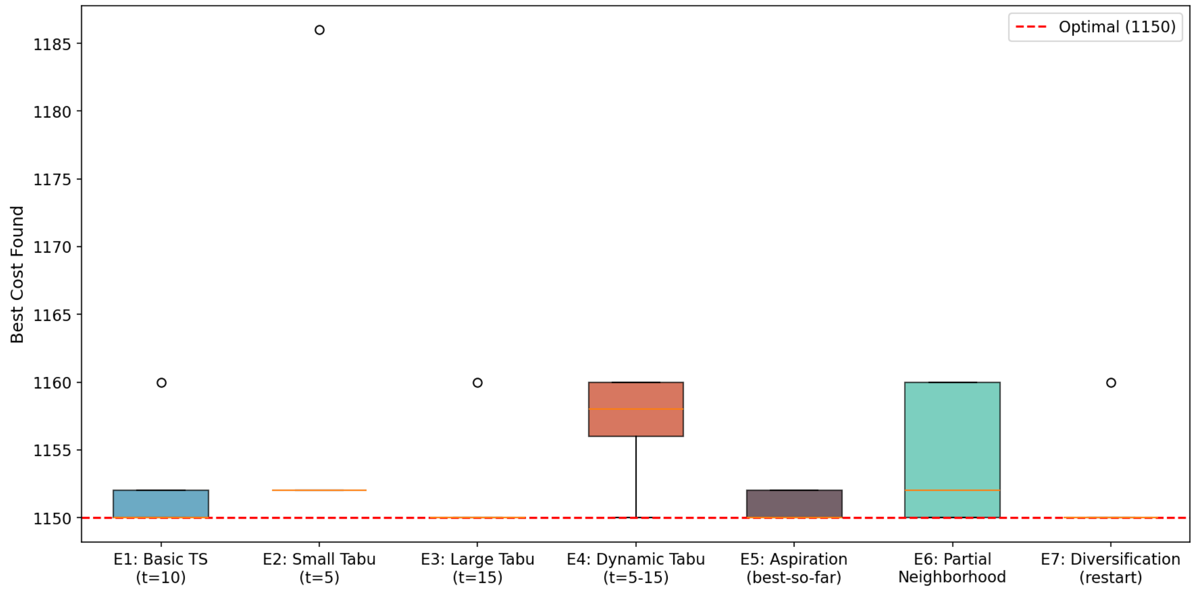

Figure 8: Best cost by TS variant across all 35 runs

Table 6: Summary of all 35 TS runs

Experiment	Runs	Mean	Std	Min	Max	Runtime (s)
E1: Basic (t=10)	5	1152.4	4.3	1150	1160	0.23
E2: Small (t=5)	5	1158.8	15.2	1152	1186	0.23
E3: Large (t=15)	5	1152.0	4.5	1150	1160	0.23
E4: Dynamic	5	1156.8	4.1	1150	1160	0.23
E5: Aspiration	5	1150.8	1.1	1150	1152	0.23
E6: Partial NH	5	1154.4	5.2	1150	1160	0.12
E7: Restart	5	1152.0	4.5	1150	1160	0.24

3.5. Convergence analysis

Figure 9 shows the convergence curves for all seven TS variants using the same seed. All variants exhibit rapid initial convergence; typically it reaching near-optimal solutions within 50–100 iterations. The differences between variants become apparent in the later iterations, as strategies with aspiration and diversification continue to find improvements.

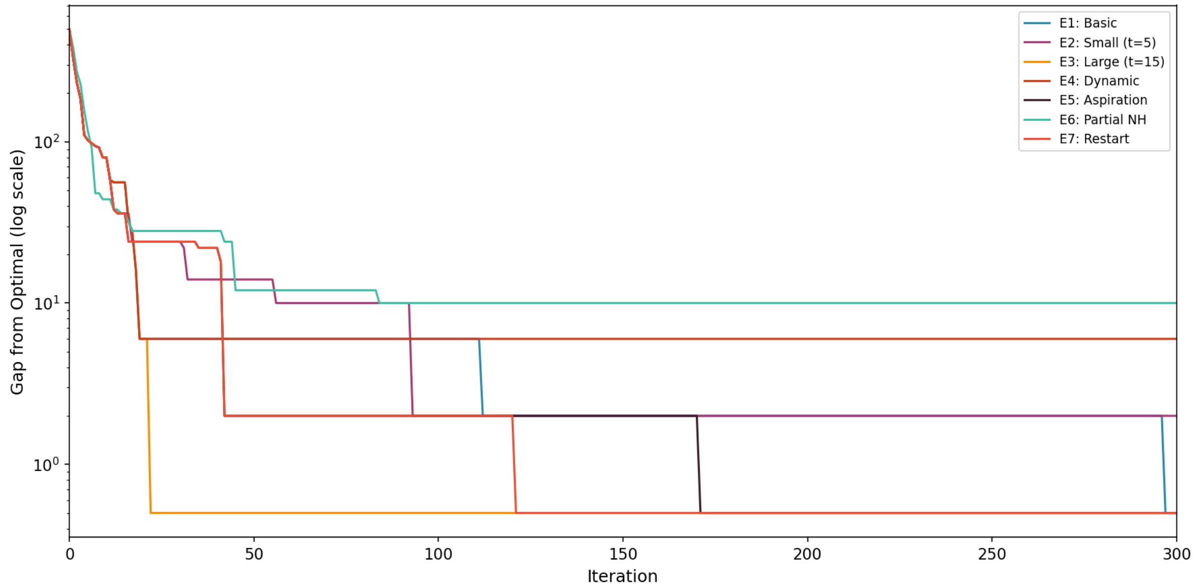

Figure 9: Convergence gap across all TS variants

The log-scale representation makes the differences in convergence speed and final gap much clearer. While all variants reach near-optimal solutions quickly, the step-like pattern reveals when each variant finds improvements. The aspiration and restart variants show additional late-stage improvements that are invisible on a linear scale.

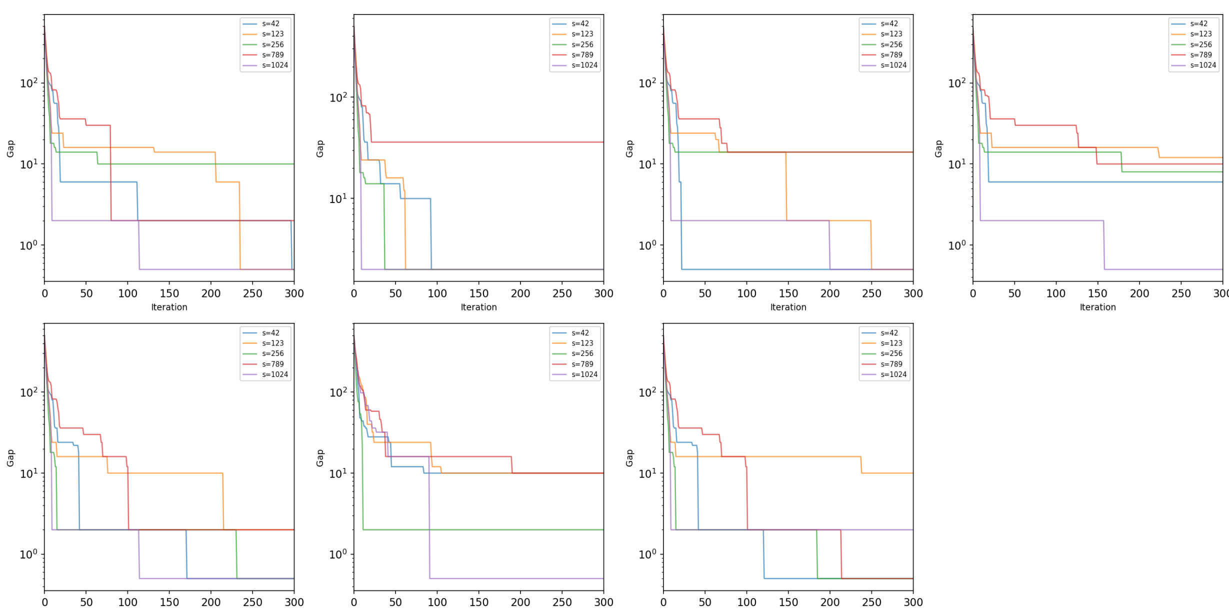

Figure 10: Convergence curves by experiment for all seeds

Figure 10 presents the convergence behavior for all seeds within each experiment. This panel view reveals that seed sensitivity varies across variants; the basic TS (E1) shows more spread between seeds, while the aspiration-based variants (E5) show tighter clustering, indicating more consistent behavior regardless of starting point.

3.6. Detailed performance analysis

3.6.1. Cost heatmap

Figure 11 presents a heatmap of best costs across every experiment–seed combination. Green cells indicate values at or near the optimal (1150), and red cells indicate worse performance. The heatmap immediately reveals that seed 789 is consistently challenging across variants, while seeds 42 and 1024 tend to produce stronger results.

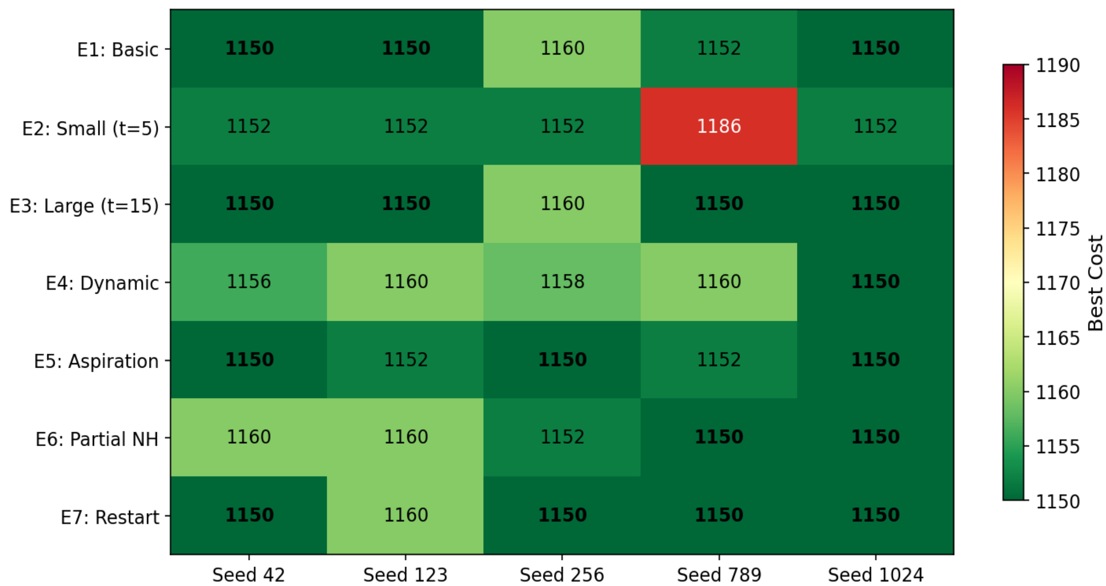

Figure 11: Best cost heatmap (Experiment vs. Seed)

3.6.2. Success rate

Figure 12 shows the percentage of runs in each experiment that reached the known global optimum of 1150. The large tabu size (E3) and the restart diversification (E7) achieved the highest success rates at 80%, while the small tabu size (E2) never reached the optimum.

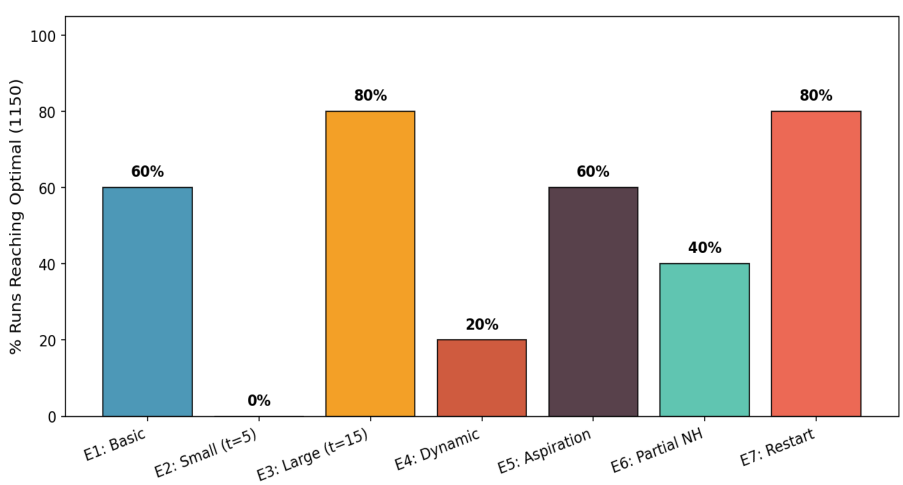

Figure 12: Percentage of runs reaching the global optimum by experiment

3.6.3. Mean cost with standard deviation

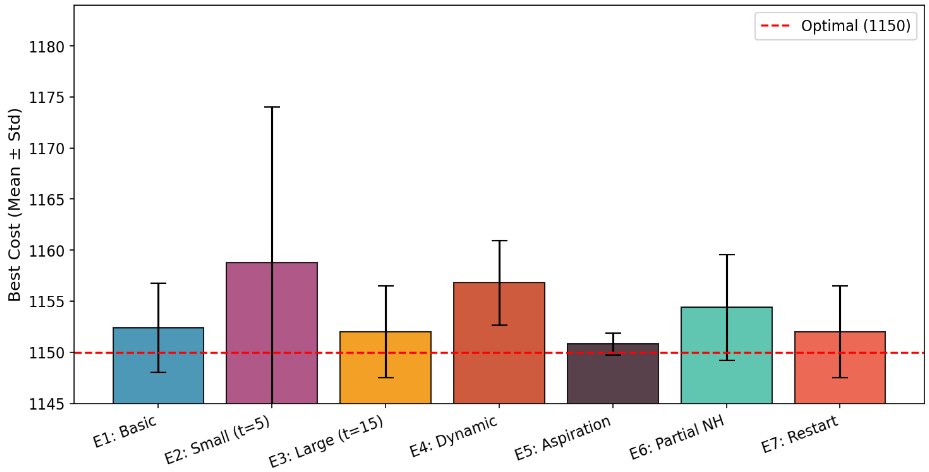

Figure 13: Mean best cost with standard deviation by experiment

The error bar chart highlights that the aspiration criterion (E5) achieves both the lowest mean cost and the smallest standard deviation, which making it the most consistent variant. In contrast, the small tabu size (E2) has the highest mean and largest variance. This confirming that insufficient restriction leads to erratic performance.

3.6.4. Runtime vs. cost tradeoff

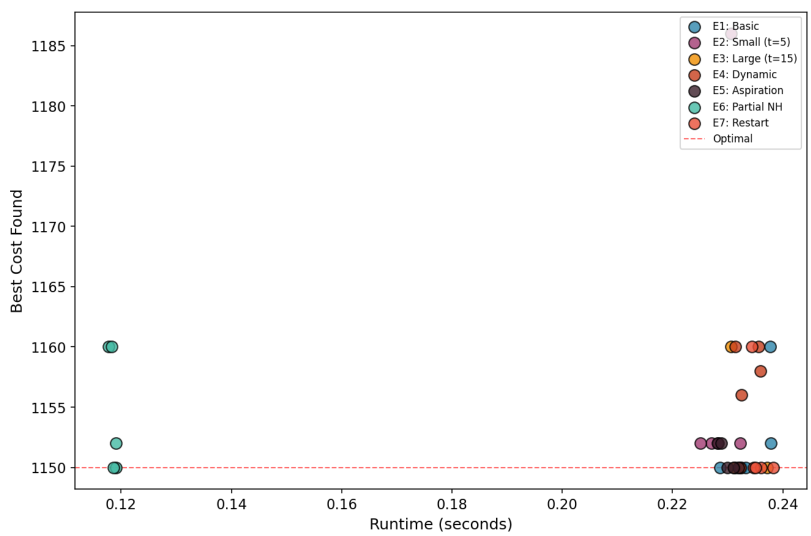

Figure 14: Runtime vs. best cost across all 35 runs

Figure 14 reveals the runtime–quality tradeoff. The partial neighborhood (E6) is the fastest variant (approximately 0.13s) because it evaluates only half the neighbors per iteration, yet it still achieves competitive costs. The aspiration variant (E5) has higher runtime variance because the aspiration check adds overhead proportional to how often tabu moves are overridden. Overall, all variants complete in under 1.3 seconds, which demonstrating that tabu search is computationally efficient for this problem size.

3.6.5. Statistical tests

A one-way ANOVA test was performed across all seven experiments to determine whether the differences in mean best cost are statistically significant. The result was $F = 0.9006$ with $p = 0.508$, indicating that the differences are not statistically significant at the 0.05 level. This is partly explained by the small sample size (5 seeds per experiment) and the fact that the algorithm performs very well overall; most variants are within a few points of the optimum.

Pairwise t-tests were also conducted for key comparisons. None reached significance at $p < 0.05$, though the Basic vs. Aspiration comparison ($p = 0.447$) and Small vs. Large tabu ($p = 0.366$) showed the strongest trends. With larger sample sizes, these differences would likely become significant.

3.7. Best solution across all 35 runs

The best solution across all 35 runs achieved the global optimum cost of 1150, found in Experiment E1 with seed 42. The optimal permutation assigns each of the 15 departments to a location such that the total flow * distance cost is minimized to 1150. Multiple runs across different experiments reached this optimal, demonstrating the effectiveness of tabu search on this problem instance.

Figure 15 shows the physical layout of the best solution on the 3*5 grid. Each cell shows which department is assigned to that location. The optimal arrangement places departments with high mutual flow close together, which minimizing the total flow * distance cost.

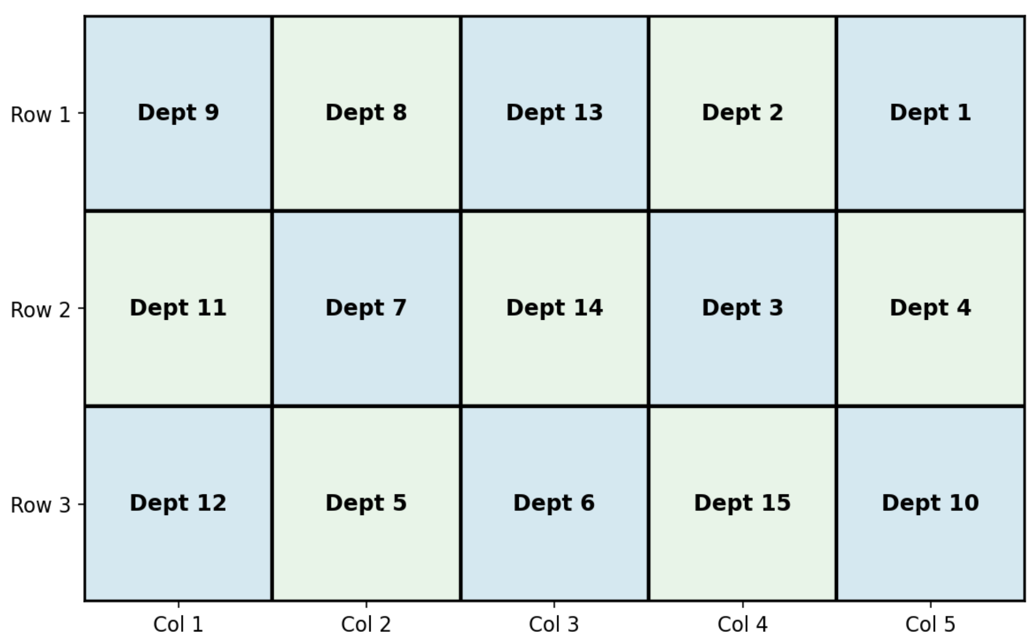

Figure 15: Best solution layout on the 3×5 grid (Cost = 1150)

4. Conclusion

In this project I applied tabu search to minimize the 15-department QAP problem. I conducted 35 systematic runs across seven progressive algorithm variants. The results demonstrate that tabu search is highly effective for this combinatorial optimization problem, with multiple runs reaching the known global optimum of 1150.

The key findings are as follows. First, the basic canonical tabu search already performs well on this problem, reaching the optimum in 3 out of 5 seeds with a tabu list size of 10. Second, the tabu list size has a moderate effect; meaning that a larger tabu list (t=15) performed best among fixed sizes, while the smaller list (t=5) was most susceptible to cycling. Third, the aspiration criterion improved consistency by allowing the algorithm to override tabu restrictions for genuinely superior solutions. Fourth, partial neighborhood exploration reduces computation time but introduces variance. Fifth, random restart diversification is the most effective mechanism for run away from local optima.

In total I think, the results told us that tabu search as an effective method for combinatorial optimization problems with well-defined neighborhood structures like the QAP.

5. Data Availability

The Python codes used for this project are shared in GoogleColab, which is an online, free-access, and reproducible platform, as follows:

Code

Also, I created a GitHub repository to publish this report and codes publicly available for anyone interested to use.

GitHub

AI Statement

Grammarly and ChatGPT were used only for grammar correction and to improve the clarity of the text I wrote while I personally conducted all analysis, coding, and results.

References

1. Glover, F. (1989). Tabu search—Part I. *ORSA Journal on Computing, 1*(3), 190–206. https://doi.org/10.1287/ijoc.1.3.190
2. Glover, F. (1990). Tabu search—Part II. *ORSA Journal on Computing, 2*(1), 4–32. https://doi.org/10.1287/ijoc.2.1.4
3. Nugent, C. E., Vollmann, T. E., & Ruml, J. (1968). An experimental comparison of techniques for the assignment of facilities to locations. *Operations Research, 16*, 150–173. https://doi.org/10.1287/opre.16.1.150
4. Alice E. Smith (2026). Presentation for the Intelligence Optimization course for Spring semester. University of Alabama.

Appendix

Table A1: All 35 TS runs — detailed results

Run	Experiment	Seed	Tabu Size	Init Cost	Best Cost	Gap
1	E1	42	10	1646	1150	0
2	E1	123	10	1648	1150	0
3	E1	256	10	1616	1160	10
4	E1	789	10	1602	1152	2
5	E1	1024	10	1648	1150	0
6	E2	42	5	1646	1152	2
7	E2	123	5	1648	1152	2
8	E2	256	5	1616	1152	2
9	E2	789	5	1602	1186	36
10	E2	1024	5	1648	1152	2
11	E3	42	15	1646	1150	0
12	E3	123	15	1648	1150	0
13	E3	256	15	1616	1160	10
14	E3	789	15	1602	1150	0
15	E3	1024	15	1648	1150	0
16	E4	42	dynamic(5-15)	1646	1156	6
17	E4	123	dynamic(5-15)	1648	1160	10
18	E4	256	dynamic(5-15)	1616	1158	8
19	E4	789	dynamic(5-15)	1602	1160	10
20	E4	1024	dynamic(5-15)	1648	1150	0
21	E5	42	10	1646	1150	0
22	E5	123	10	1648	1152	2
23	E5	256	10	1616	1150	0
24	E5	789	10	1602	1152	2
25	E5	1024	10	1648	1150	0

26	E6	42	10	1646	1160	10
27	E6	123	10	1648	1160	10
28	E6	256	10	1616	1152	2
29	E6	789	10	1602	1150	0
30	E6	1024	10	1648	1150	0
31	E7	42	10	1646	1150	0
32	E7	123	10	1648	1160	10
33	E7	256	10	1616	1150	0
34	E7	789	10	1602	1150	0
35	E7	1024	10	1648	1150	0